MP3 Performance Analysis

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L08: Numerical Accuracy

MP3 M1: Trajectory Divergence Map

Region: x ∈ [-0.7530, -0.7490], y ∈ [0.0990, 0.1030]

Threshold $\tau = 0.01$

Max iterations = 1000

Observation:

The left side of the image has low values. These are points inside or far outside the Mandelbrot set. Here, float32 and float64 give the same result.

The right side (Seahorse Valley) has high values. This is the fractal edge. Here, float32 and float64 take many iterations before they start to differ

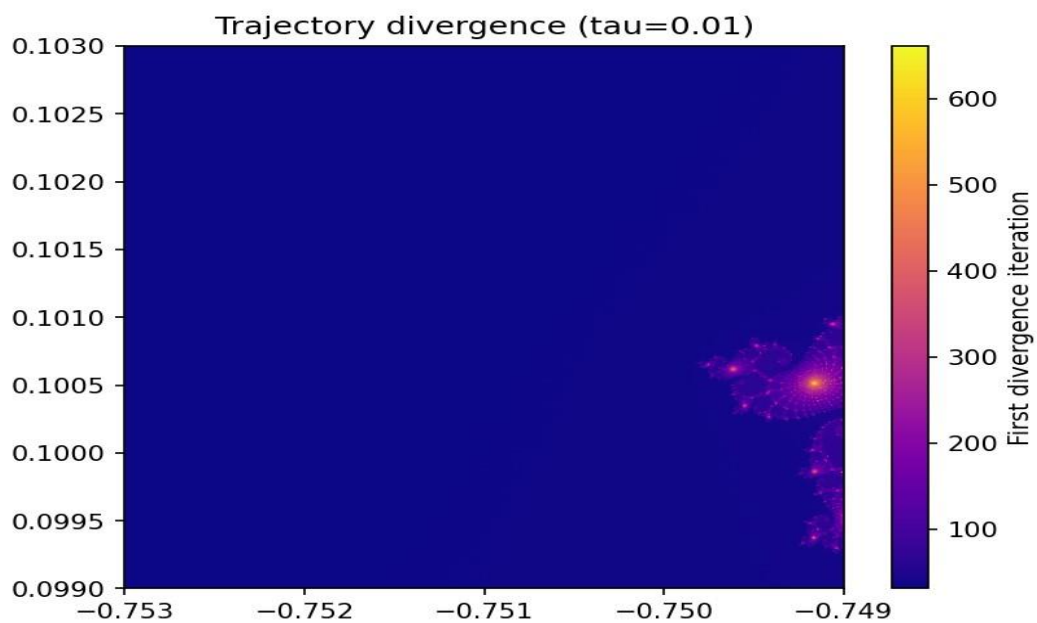

Figure 1: Trajectory divergence map (float32 vs float64) at Seahorse Valley

MP3 M2: Sensitivity Map

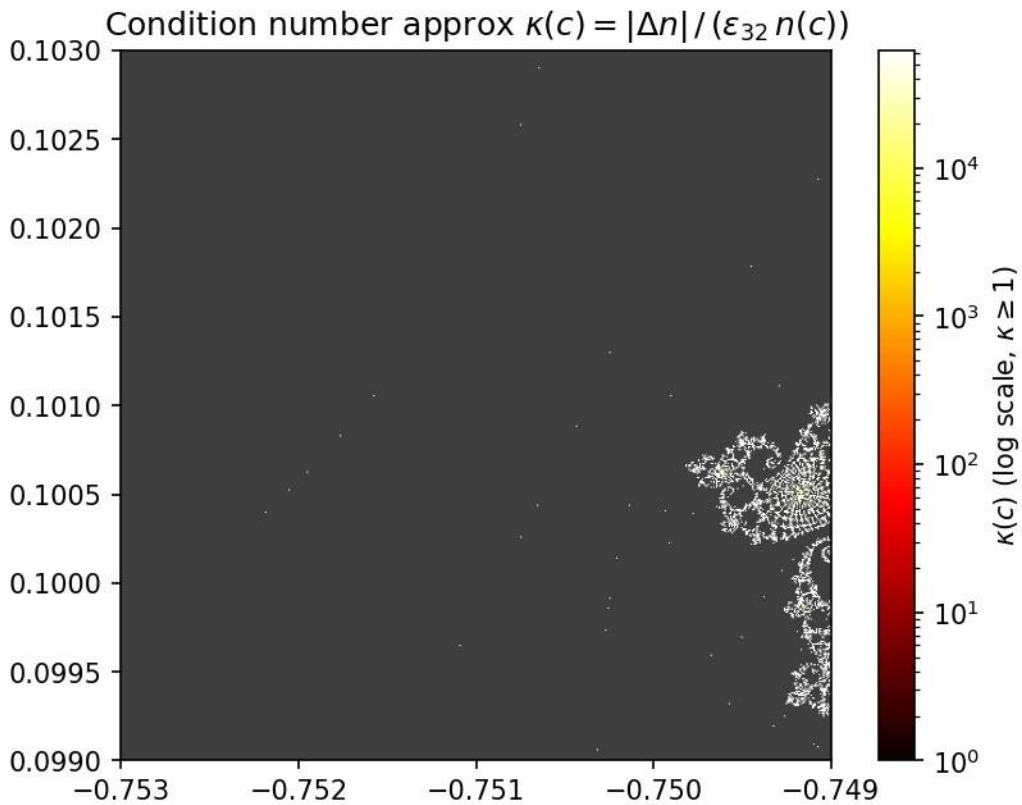

Figure 2: Sensitivity map (condition number approximation) at Seahorse Valley

Observation:

Areas with low values are stable. Small changes in input do not change the output much.

Areas with high values are near the fractal edge. Here, even a tiny change in input causes a big change in output.

L09: Testing and Documentation

L09.M1: Test Suite
pytest -v: 5 tests passed
Coverage for test_mandelbrot.py: 100%
Overall project coverage: 6%

L09.M2: Docstrings and Type Hints

All public functions have NumPy-style docstrings
Type hints added to all parameters and return values
ruff check: 0 errors (All checks passed)

L10: GPU Performance

MP3 M1: GPU Mandelbrot float32

Device: Intel(R) HD Graphics 620
Grid size: 1024×1024
Max iterations: 100
Runtime: 0.831 seconds

MP3 M2: Float32 vs Float64 Comparison
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float32 runtime: 0.831 seconds
float64 runtime: 0.717 seconds
Ratio (f64/f32): 0.86x
Max pixel difference: 37 pixels

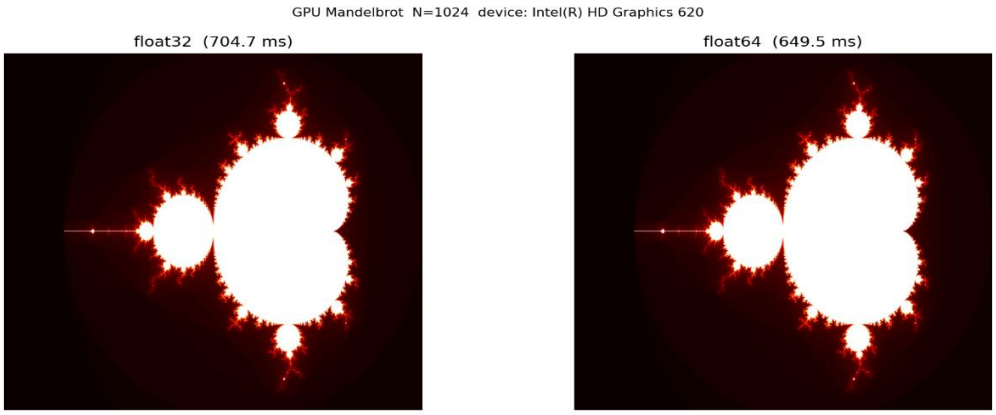

Figure 3: GPU Mandelbrot output (float32 left, float64 right)

MP3 M3: Combined Benchmark

Performance Table

Implementation	Resolution	Runtime (s)
Naive Python	1024×1024	10.142
NumPy	1024×1024	1.603
Numba f32	1024×1024	0.094
Multiprocessing	1024×1024	0.056
Dask Local	1024×1024	0.080
Dask Cluster	4096×4096	0.107
GPU f32	1024×1024	0.831
GPU f64	1024×1024	0.717

Bar chart (log scale)

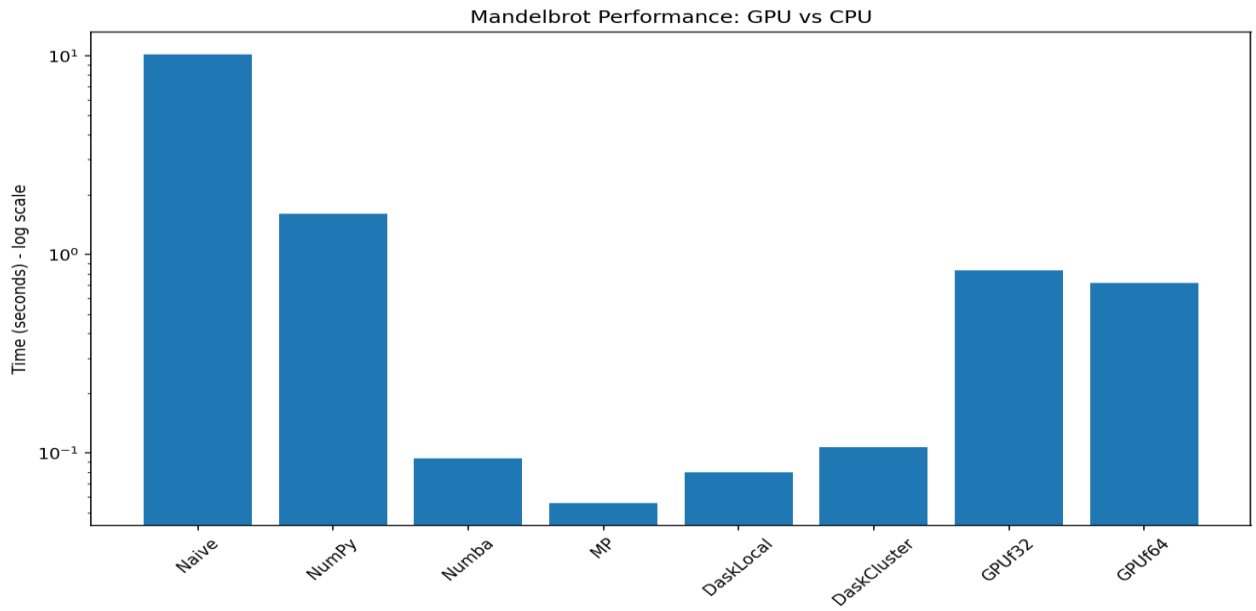

Figure 4: Performance comparison of all implementations (log scale)